

Skills Summary

Separable Equations: the three places it goes wrong

Use this reference while completing your worksheet or when studying.

Skill 1 — Separating before integrating

Two questions, in this order. (1) Does the right side factor as (something in x) times (something in y)? A sum is not a veto: $x + xy = x(1 + y)$ separates, $x + y$ does not. (2) Only after every y sits with dy may you integrate — and then each side is integrated in *its own* variable, with the power rule dividing by the new exponent.

Example: A student integrates $\int 6x dx$ as $6x^2$. Repair the general solution of $\frac{dy}{dx} = 6xy$.

Step 1. The separation is fine, so the only thing to check is the antiderivative: the power rule raises the exponent and then divides by the new one, which the student skipped.

Step 2. $\int \frac{dy}{y} = \int 6x dx$ gives $\ln |y|$ on the left and $\frac{6}{2}x^2 = 3x^2$ on the right.
 $\Rightarrow \ln |y| = 3x^2 + C$

Try it: Find the general solution of $\frac{dy}{dx} = 10xy$.

check: $\ln |y| = 5x^2 + C$

Skill 2 — Where the constant belongs

The constant moves when you exponentiate. From $\ln |y| = F(x) + C$ you get $y = e^{F(x)+C} = Ae^{F(x)}$: the constant *multiplies*. Writing $y = e^{F(x)} + C$ is the single most common slip in this topic. For a positive starting value, A is just y at the starting point — no extra algebra needed.

Example: Solve $\frac{dy}{dt} = 0.02y$ with $y(0) = 500$, then find $y(50)$.

Step 1. The equation is exponential growth, so the answer has the form Ae^{kt} ; the job is to place the constant as a multiplier and read A off the starting value.

Step 2. $\ln |y| = 0.02t + C$ gives $y = Ae^{0.02t}$, and $y(0) = 500$ forces $A = 500$, so $y = 500e^{0.02t}$ and $y(50) = 500e^1$.

$\Rightarrow y(50) \approx 1359.141$

Try it: Solve $\frac{dy}{dt} = 0.03y$ with $y(0) = 200$, then find $y(100)$.

check: $y(100) \approx 200e^3$

Skill 3 — Substituting the starting values

Substitute into the line you actually wrote. Put the starting pair into the *implicit* equation, exactly as it stands, before rearranging. If that line reads $\frac{1}{2}y^2 = \dots + C$, the constant comes from $\frac{1}{2}y_0^2$, not from y_0 or y_0^2 . Swapping the two coordinates, or reading the constant off a numerator, are the two ways this line fails.

Example: Solve $\frac{dy}{dx} = \frac{6x}{y}$ with $y(0) = 4$ and $y > 0$; find $y(3)$.

Step 1. The implicit form will be quadratic in y , so substitute the starting pair into that quadratic form and solve for the constant before taking any square root.

Step 2. $y \, dy = 6x \, dx$ gives $\frac{1}{2}y^2 = 3x^2 + C$; at $(0, 4)$ this reads $8 = C$, so $y^2 = 6x^2 + 16$ and $y(3) = \sqrt{54 + 16}$.

$$\Rightarrow \mathbf{y(3) \approx 8.367}$$

Try it: Solve $\frac{dy}{dx} = \frac{8x}{y}$ with $y(0) = 3$ and $y > 0$; find $y(2)$.

check: $y(2) \approx 6.403$